10.5 Bond parameters

A covalent bond is characterised by parameters such as bond length, bond angle, bond order etc... A brief description of some of the bond parameters is given below.

10.5.1 Bond length

The distance between the nuclei of the two covalently bonded atoms is called bond length. Consider a covalent molecule A-B. The bond length is given by the sum of the radii of the bonded atoms ($r_A + r_B$). The length of a bond can be determined by spectroscopic, x-ray diffraction and electron-diffraction techniques The bond length depends on the size of the atom and the number of bonds (multiplicity) between the combining atoms.

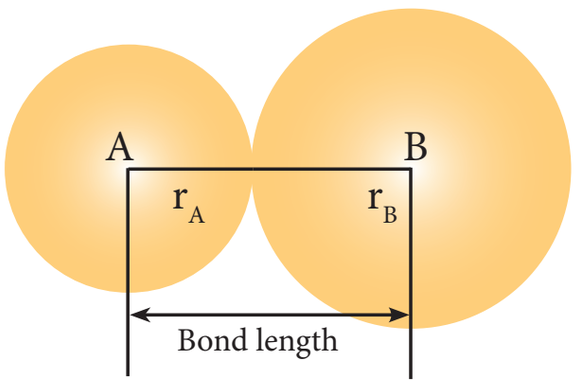

Fig 10. 13 Bond length of covalent molecule A–B

Greater the size of the atom, greater will be the bond length. For example, carbon-carbon single bond length (1.54 Å) is longer than the carbon-nitrogen single bond length (1.43 Å).

Increase in the number of bonds between the two atoms decreases the bond length. For example, the carbon-carbon single bond is longer than the carbon-carbon double bond (1.33 Å) and the carbon-carbon triple bond (1.20 Å).

10.5.2 Bond order

The number of bonds formed between the two bonded atoms in a molecule is called the bond order. In Lewis theory, the bond order is equal to the number of shared pair of electrons between the two bonded atoms. For example in hydrogen molecules, there is only one shared pair of electrons and hence, the bond order is one. Similarly, in H_2O, HCl, Methane, etc the central atom forms single bonds with bond order of one.

Table 10.2 Bond order of some common bonds:

S. No.	Molecule	Bonded atoms	Bond order (No. of shared pair of electrons between bonded atoms)
1	H_2	H-H	1
2	O_2	O=O	2
3	N_2	N≡N	3
4	HCN	C≡N	3
5	HCHO	C=O	2
6	CH_4	C–H	1
7	C_2H_4	C=C	2

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